# 📝 Unit 8 Guided Notes — Transformations

Slide it. Flip it. Turn it. Resize it. Then describe what changed and what stayed the same.

*Standards: NC.8.G.1 · NC.8.G.2 · NC.8.G.3 · NC.8.G.4*

A **transformation** is a rule that moves every point of a figure in a predictable way. In this unit you'll learn four of them — **translations** (slides), **reflections** (flips), **rotations** (turns), and **dilations** (resizes) — and you'll classify the result as **congruent** (same size and shape) or **similar** (same shape, different size). Every transformation in this unit happens on the coordinate plane, so you'll be reading coordinates, applying a rule, and plotting the image. Have your pencil and grid ready.

# 1 Transformation Vocabulary

Before we slide, flip, turn, or resize anything, we need shared language. Every transformation moves points by a rule — and we always describe what changed and what stayed the same.

📌 **Transformation:** A rule that moves, flips, turns, or resizes every point of a figure in a predictable way.

📌 **Pre-image:** The *original* figure, BEFORE the transformation. Vertices use plain letters: A, B, C.

📌 **Image:** The *new* figure, AFTER the transformation. Vertices use a prime mark: A′, B′, C′ (read "A prime").

### Two Families of Transformations

| Family | What It Preserves | Types | Result |
|---|---|---|---|
| **Rigid** (Isometry) | Size AND shape | Translation, Reflection, Rotation | Pre-image & image are **CONGRUENT** |
| **Non-Rigid** | Shape only (size changes) | Dilation | Pre-image & image are **SIMILAR** |

💡 **Big idea:** If size stays the same, it's **congruent**. If size changes but shape stays the same, it's **similar**.

### Quick Check

**Q1** **A triangle is reflected over the y-axis. The pre-image and image are…**

✓ Reflection is a rigid transformation — size and shape are preserved, so the figures are congruent.

**Q2** **A square is dilated by a scale factor of 3. The pre-image and image are…**

✓ Dilation is non-rigid — the shape stays the same but size changes, so the figures are similar but not congruent.

# 2 Translations (Slides)

📌 **Translation:** Every point slides the *same distance* in the *same direction*. The shape is NOT rotated or flipped — it just moves.

## The Rule

**(x, y) → (x + h, y + k)**

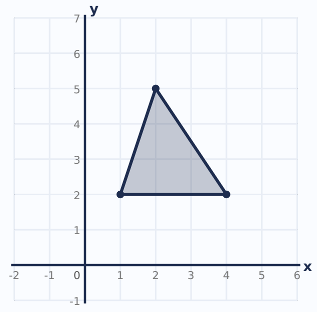
**h** = horizontal shift | **+** moves right, **−** moves left
**k** = vertical shift | **+** moves up, **−** moves down

💡 **Tip:** Add the rule's numbers to the matching coordinates. The signs do the work — no need to memorize "right vs. left."

## Worked Example — Click each step to reveal

**Example 1: Triangle ABC has vertices A(1, 2), B(4, 2), C(2, 5). Translate by (x − 3, y + 1).**

*Pre-image △ABC*




**Step 1 — Identify h and k:** h = −3 (shift left 3), k = +1 (shift up 1)

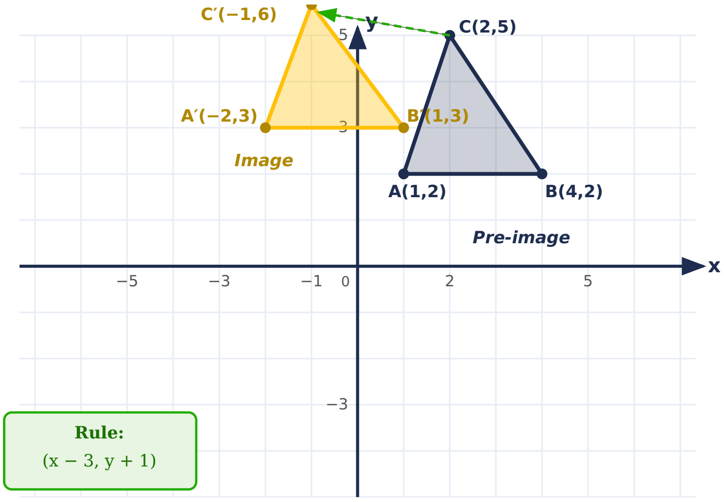
**Step 2 — Apply rule to A(1, 2):** (1 + (−3), 2 + 1) = (−2, 3) → A′(−2, 3)

**Step 3 — Apply rule to B(4, 2):** (4 + (−3), 2 + 1) = (1, 3) → B′(1, 3)

**Step 4 — Apply rule to C(2, 5):** (2 + (−3), 5 + 1) = (−1, 6) → C′(−1, 6)

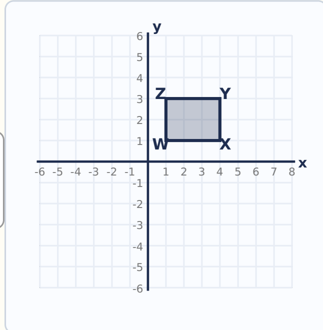
✓ **Image vertices:** A′(−2, 3), B′(1, 3), C′(−1, 6)

## Visual



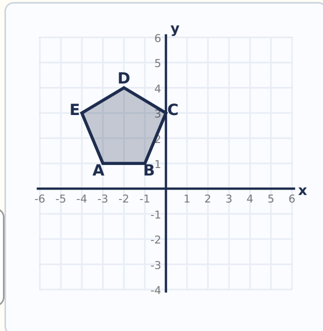

## Practice

**P1** **Rectangle WXYZ has vertices W(1, 1), X(4, 1), Y(4, 3), Z(1, 3). Translate the rectangle by the rule (x + 2, y − 4). What are the coordinates of W′?**




✓ Apply (x + 2, y − 4) to W(1, 1): (1 + 2, 1 − 4) = (3, −3). Every other vertex shifts the same way.

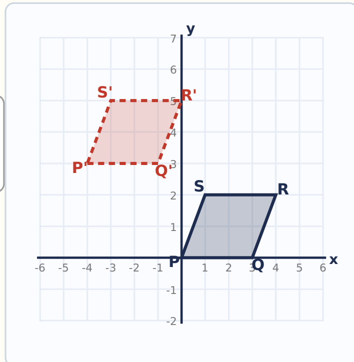
**P2** **Pentagon ABCDE has vertex A(−3, 1). The whole pentagon is translated by (x + 5, y − 2). Type the coordinates of A′.**




**A′ = (** x **,** y **)**

✓ A(−3, 1) → A′(−3 + 5, 1 − 2) = (2, −1). All five vertices shift by the same rule.

**P3** **Parallelogram PQRS was translated to P′Q′R′S′ (shown). P(0, 0) maps to P′(−4, 3). Which rule was used?**




✓ P(0, 0) → P′(−4, 3) means x decreased by 4 and y increased by 3. Rule: (x − 4, y + 3).

# 3 Reflections (Flips)

📌 **Reflection:** Every point flips across a **line of reflection** (the mirror line). Each point lands the same distance from the mirror, on the opposite side.

📌 **NC Standard 8.G.3:** 8th grade focuses on reflections across the **x-axis** and **y-axis**.

💡 **Beyond the x- and y-axis:** A reflection can happen across *any* line — not just the x- and y-axis. In high school you'll work with reflections over lines like **y = x**, **y = −x**, **y = 3** (a horizontal line), or **x = −2** (a vertical line). The big idea is the same every time: each point lands the same distance from the mirror, on the opposite side. We'll stick with the x- and y-axis this year so you can master the foundation.

### The Rules

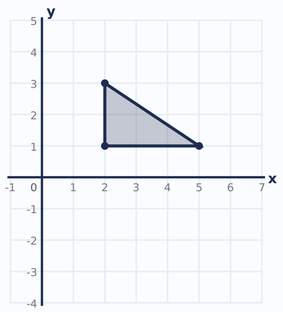
| Line of Reflection | Rule | Example: (3, 4) → | What Changes |
|---|---|---|---|
| **x-axis** | `(x, y) → (x, −y)` | (3, −4) | y flips sign |
| **y-axis** | `(x, y) → (−x, y)` | (−3, 4) | x flips sign |

📌 **Memory trick:** The coordinate that *matches the axis name* is the one that **stays the same**. Reflect over the x-axis → x stays. Reflect over the y-axis → y stays.

### Worked Example — Click each step to reveal

**Example 2: Reflect triangle DEF with D(2, 3), E(5, 1), F(2, 1) over the x-axis.**

*Pre-image △DEF (mirror = x-axis)*




**Step 1 — Identify the rule:** Reflection over x-axis: (x, y) → (x, −y). Keep x; flip the sign of y.

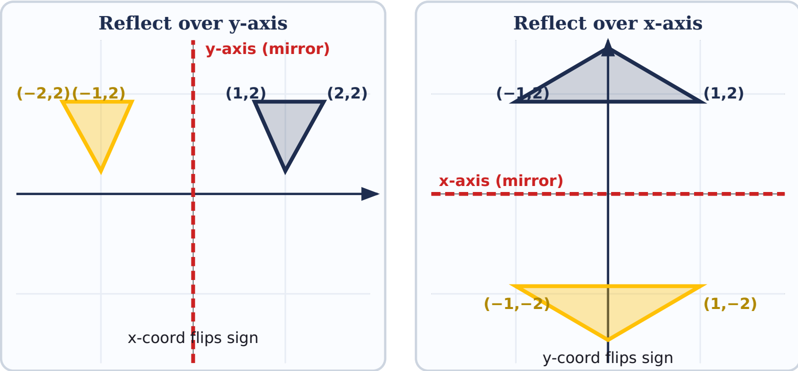
**Step 2 — Apply to D(2, 3):** (2, −3) → D′(2, −3)

**Step 3 — Apply to E(5, 1):** (5, −1) → E′(5, −1)

**Step 4 — Apply to F(2, 1):** (2, −1) → F′(2, −1)

✓ **Image vertices:** D′(2, −3), E′(5, −1), F′(2, −1)

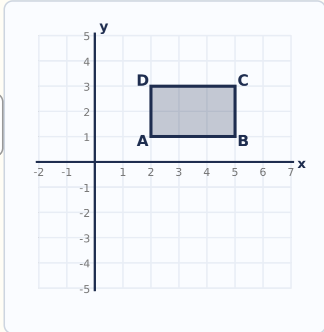
## Visual — Two Mirrors Side by Side




## Practice

**P1** **Rectangle ABCD has vertex C(5, 3). The whole rectangle is reflected over the x-axis. What are the coordinates of C′?**

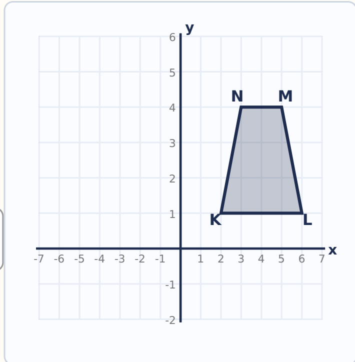
✓ Reflect over the x-axis: (x, y) → (x, −y). C(5, 3) → C′(5, −3).




**P2** **Trapezoid KLMN has vertex M(5, 4). The whole trapezoid is reflected over the y-axis. Type the coordinates of M′.**

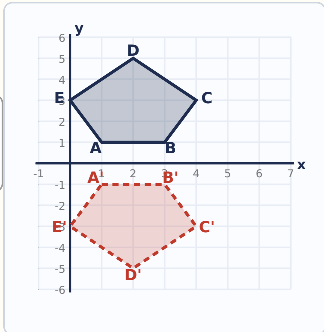
**M′ = (** x **,** y **)**

✓ Reflect over the y-axis: (x, y) → (−x, y). M(5, 4) → M′(−5, 4).




**P3** **Pentagon ABCDE was reflected to A′B′C′D′E′ (shown). Which line is the mirror (line of reflection)?**

✓ Each image point has the same x but the opposite y (e.g., A(1, 1) → A′(1, −1)). That's the rule for reflection over the x-axis.




4 Rotations (Turns)

## 4 Rotations (Turns)

📌 **Rotation:** Every point turns a fixed angle around a **center of rotation**. In this unit, the center is always the **origin (0, 0)**.

📌 **NC Standard 8.G.3:** 8th grade focuses on rotations of **90°, 180°, and 270°** about the origin.

↻ **Direction matters — always read carefully:** A rotation can be **counterclockwise (CCW)** or **clockwise (CW)**. Both are valid and you'll see both on assignments and tests — *always check which direction the problem is asking for*. Helpful link: 90° CW = 270° CCW, and 270° CW = 90° CCW. 180° is the same either direction.

### The Rules (About the Origin)

Two columns of rules — one for counterclockwise (CCW), one for clockwise (CW). Same angles, opposite directions.

| Rotation | Counterclockwise (CCW) | Clockwise (CW) | Example: (3, 4) → CCW | Example: (3, 4) → CW |
|---|---|---|---|---|
| **90°** | `(x, y) → (−y, x)` | `(x, y) → (y, −x)` | (−4, 3) | (4, −3) |
| **180°** | `(x, y) → (−x, −y)` | `(x, y) → (−x, −y)` | (−3, −4) | (−3, −4) |
| **270°** | `(x, y) → (y, −x)` | `(x, y) → (−y, x)` | (4, −3) | (−4, 3) |

📌 **Memory pattern:**
- **90° CCW:** swap, then negate the new x | **90° CW:** swap, then negate the new y
- **180°:** keep order, negate both (same either direction)
- **270° CCW:** swap, then negate the new y | **270° CW:** swap, then negate the new x

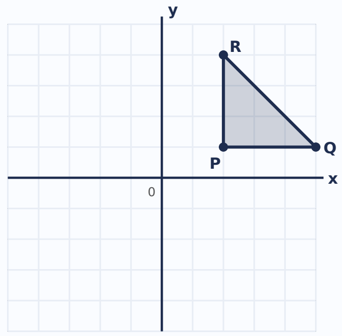
🔗 **Notice the symmetry:** 90° CCW and 270° CW share the same rule. 270° CCW and 90° CW share the same rule. That's because going 90° one way takes you to the exact same spot as going 270° the other way.

⚠ **Watch out:** 180° is the only rotation where the coordinates are NOT swapped. For 90° and 270°, you must swap x and y first, THEN apply the sign rule.

### Worked Example — Click each step to reveal

**Example 3: Rotate triangle PQR with P(2, 1), Q(5, 1), R(2, 4) 90° counterclockwise about the origin.**

*Pre-image △PQR*




**Step 1 — Identify the rule:** 90° CCW about origin: (x, y) → (−y, x)

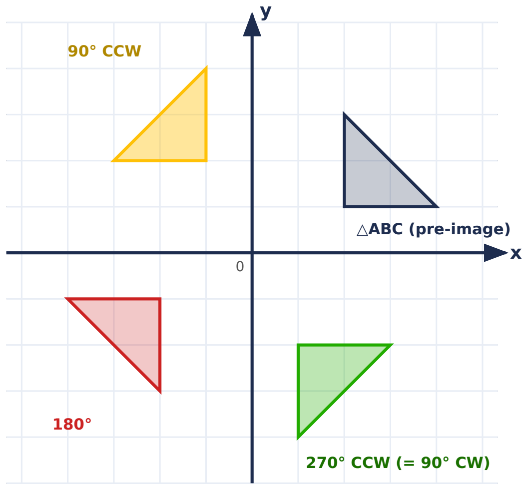
**Step 2 — Apply to P(2, 1):** (−1, 2) → P′(−1, 2)

**Step 3 — Apply to Q(5, 1):** (−1, 5) → Q′(−1, 5)

**Step 4 — Apply to R(2, 4):** (−4, 2) → R′(−4, 2)

✓ **Image vertices:** P′(−1, 2), Q′(−1, 5), R′(−4, 2)

## Visual — One Triangle, Three Rotations

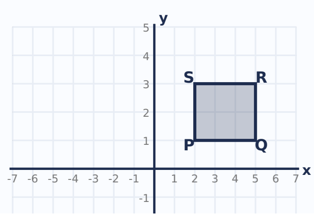


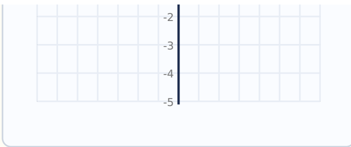

*Same triangle △ABC, three different rotations. Notice the triangle stays the same size and shape — only its position and orientation change.*

## Practice

**P1** **Rectangle PQRS has vertex R(5, 3). The whole rectangle is rotated 180° about the origin. What are the coordinates of R′?**

✓ 180° rule: (x, y) → (−x, −y). R(5, 3) → R′(−5, −3). Both signs flip.

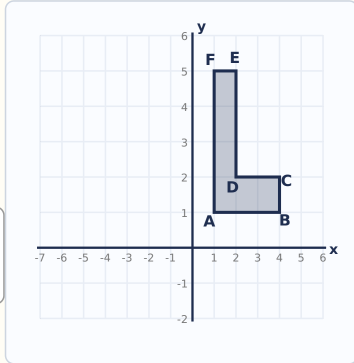




**P2** **L-shape ABCDEF has vertex E(2, 5). The whole shape is rotated 90° counterclockwise (CCW) about the origin. Type the coordinates of E′.**

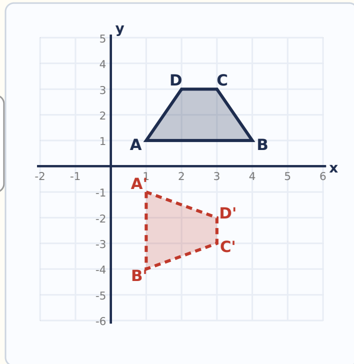
**E′ = (** x **,** y **)**

✓ 90° CCW rule: (x, y) → (−y, x). E(2, 5) → E′(−5, 2). Every vertex follows the same rule.




**P3** **Trapezoid ABCD was rotated to A′B′C′D′ (shown). All rotations are about the origin. Which rotation was used?**

✓ Check A(1, 1) → A′(1, −1). The rule (x, y) → (y, −x) is 90° CW. Also matches B(4, 1) → B′(1, −4).




## 5 Dilations (Resize)

📌 **Dilation:** A transformation that **resizes** a figure by a **scale factor k** from a center point (usually the origin). The shape stays the same; the size changes.

📌 **NC Standard 8.G.4:** Dilations in 8th grade are centered at the origin. Scale factor k can be a whole number, decimal, or fraction.

### The Rule (Centered at the Origin)

$$(x,\ y) \rightarrow (kx,\ ky)$$

Multiply *both* coordinates by the scale factor **k**.

### What k Tells You

| Scale Factor k | Effect | Image vs. Pre-image |
|---|---|---|
| **k > 1** | Enlargement | Image is **bigger** |

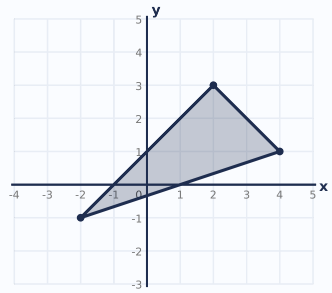
| | | |
|---|---|---|
| **k = 1** | No change | Same size (identity) |
| **0 < k < 1** | Reduction | Image is **smaller** |

⚠ **Watch out:** Dilation is the ONLY transformation in this unit that is NOT rigid. The pre-image and image are SIMILAR, but NOT congruent (unless k = 1).

### Worked Example — Click each step to reveal

**Example 4: Triangle XYZ has X(2, 3), Y(4, 1), Z(−2, −1). Dilate by scale factor k = 3 (centered at origin).**

*Pre-image △XYZ*




**Step 1 — Identify the rule:** (x, y) → (3x, 3y). Multiply both coordinates by 3.

**Step 2 — Apply to X(2, 3):** (3 · 2, 3 · 3) = (6, 9) → X′(6, 9)

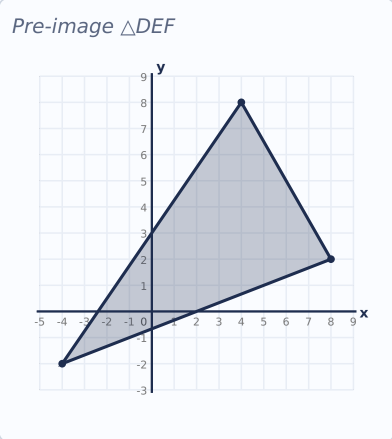
**Step 3 — Apply to Y(4, 1):** (3 · 4, 3 · 1) = (12, 3) → Y′(12, 3)

**Step 4 — Apply to Z(−2, −1):** (3 · (−2), 3 · (−1)) = (−6, −3) → Z′(−6, −3)

✓ **Image vertices:** X′(6, 9), Y′(12, 3), Z′(−6, −3) — bigger by a factor of 3

### Worked Example — Reduction

**Example 5: Triangle DEF has D(4, 8), E(8, 2), F(−4, −2). Dilate by scale factor k = 1/2 (centered at origin).**




**Step 1 — Rule:** (x, y) → ((1/2)x, (1/2)y). Multiply both coordinates by 1/2 (or divide by 2).

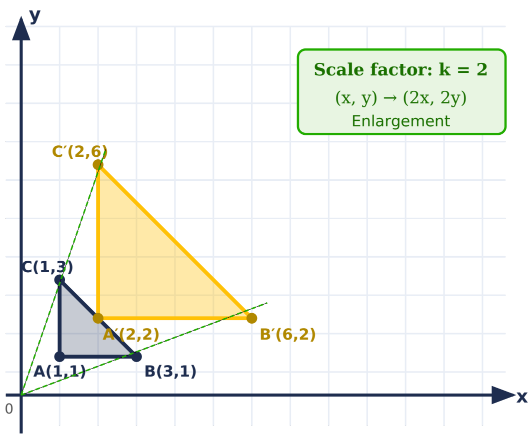
**Step 2 — Apply to D(4, 8):** ((1/2)(4), (1/2)(8)) = (2, 4) → D′(2, 4)

**Step 3 — Apply to E(8, 2):** ((1/2)(8), (1/2)(2)) = (4, 1) → E′(4, 1)

**Step 4 — Apply to F(−4, −2):** ((1/2)(−4), (1/2)(−2)) = (−2, −1) → F′(−2, −1)

✓ **Image vertices:** D′(2, 4), E′(4, 1), F′(−2, −1). Since 0 < k < 1, this is a reduction — the image is half the size of the pre-image.

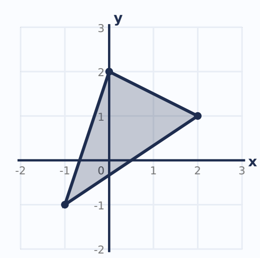
## Visual — Dilation by k = 2



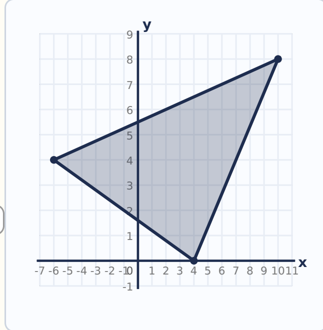

## Practice

**P1** **Triangle ABC has vertices A(2, 1), B(0, 2), C(−1, −1). The triangle is dilated by scale factor k = 4 (centered at origin). What are the new coordinates of A′?**

✓ Multiply both coordinates by 4: A′ = (2 · 4, 1 · 4) = (8, 4). Apply the same rule to B and C: B′(0, 8), C′(−4, −4).



**P2** **Triangle ABC has vertices A(10, 8), B(4, 0), C(−6, 4). The triangle is dilated by k = 1/2 (centered at origin). Type the new coordinates of vertex A′:**




**A′ = (** x **,** y **)**

**P3** **A figure is dilated and the image is *smaller* than the pre-image. The scale factor k must be...**

✓ A scale factor between 0 and 1 (like 1/2 or 0.4) produces a reduction — the image is smaller.

## 6 Congruence & Similarity

Every transformation we've studied tells us how two figures are related. Here's the big-picture connection:

📌 **Congruent figures (≅):** Same size AND same shape. One can be mapped to the other using only **rigid transformations** (translations, reflections, rotations — or any combination).

📌 **Similar figures (~):** Same shape but possibly different size. One can be mapped to the other using **rigid transformations AND a dilation**.

**The Rule**

If you can get from Figure A to Figure B using **only translations, reflections, and rotations** → they are **CONGRUENT**.

If you need a **dilation** somewhere in the sequence → they are **SIMILAR** (but not congruent, unless k = 1).

### What Gets Preserved?

| Property | Translations | Reflections | Rotations | Dilations |
|---|---|---|---|---|
| **Side lengths** | ✓ same | ✓ same | ✓ same | ✗ scaled by k |
| **Angle measures** | ✓ same | ✓ same | ✓ same | ✓ same |
| **Shape** | ✓ same | ✓ same | ✓ same | ✓ same |
| **Size** | ✓ same | ✓ same | ✓ same | ✗ changes |

💡 **Notice:** Dilations are the only transformation that changes *side lengths*. Angles stay the same in every transformation — that's why the shape never distorts.

### Worked Example — Click each step to reveal

**Example 6: Triangle ABC is reflected over the y-axis, then translated by (x + 2, y − 1). Are the pre-image and image congruent, similar, or neither?**

**Step 1 — List the transformations used:** Reflection over y-axis (rigid) + Translation (rigid).

**Step 2 — Any dilations?** No — both transformations are rigid (no resizing).

✓ **Conclusion:** The figures are CONGRUENT. Side lengths and angles are all preserved.

**Example 7: Triangle PQR is dilated by k = 2, then rotated 90° CCW. Congruent, similar, or neither?**

**Step 1 — List the transformations:** Dilation by k = 2 (non-rigid) + Rotation (rigid).

**Step 2 — Was a dilation used (with k ≠ 1)?** Yes — the dilation changes side lengths by a factor of 2.

✓ **Conclusion:** The figures are SIMILAR but NOT congruent. Same shape, different size.

### Practice

**P1** **Triangle A has side lengths 3, 4, 5. Triangle B has side lengths 6, 8, 10 and the same angles. Triangles A and B are…**

✓ Same shape, but B's sides are 2× A's sides. They are similar (related by a dilation with k = 2) but NOT congruent.

**P2** **A square is translated 5 units right, then rotated 180° about the origin. The pre-image and image are…**

✓ Both transformations are rigid (no dilation). Size and shape are preserved → congruent.

**P3** **Which transformation preserves angle measure but NOT side length?**

✓ Dilation is the only one that resizes the figure (changes side lengths), while angles remain the same.

**⭐ Challenge** **A pentagon undergoes a sequence: reflection over the x-axis, dilation by k = 3, then rotation 90° CCW. The pre-image and image are…**

✓ The dilation by k = 3 changes the side lengths, so they're not congruent. But all angles and proportions are preserved → similar.

## ✓ Unit 8 Wrap-Up

You've now seen all four transformation types and how they connect to congruence and similarity. Quick recap:

- **Translation:** (x, y) → (x + h, y + k) — slide. Rigid.
- **Reflection over x-axis:** (x, y) → (x, −y) — y flips. Rigid.
- **Reflection over y-axis:** (x, y) → (−x, y) — x flips. Rigid.
- **Rotation 90° CCW:** (x, y) → (−y, x). Rigid.
- **Rotation 180°:** (x, y) → (−x, −y). Rigid.
- **Rotation 270° CCW:** (x, y) → (y, −x). Rigid.
- **Dilation (k from origin):** (x, y) → (kx, ky). Non-rigid.

💡 **Next step:** Head to the Unit 8 page to try the practice problems and the Transformation Quest game.

← Back to Unit 8